

Ku/C-band dish antenna receives the information and sends it to the receiver, which decodes and saves it to its memory.

The receiver then shares this information with other devices using Wi-Fi. Wi-Fi-enabled devices like laptops and smartphones connect to the receiver access point and open any browser. Information is then displayed on the screen.

The steps are shown in detail in Figs 1a through 1d.

Reception of the Outernet

Currently, reception of the Outernet is possible from the following three devices:

- Lighthouse, which is a small receiver that looks like a set-top box
- Lantern, which is currently under development in the Outernet with Indiegogo campaign
- Build-your-own receiver using USB satellite tuner and Raspberry Pi

Lighthouse. This is a small receiver that looks like set-top box. It receives one-way data multicast from the Outernet network of six geostationary satellites. It requires a satellite dish for signal reception. Lighthouse is perfect for places where there is no Internet access, is too expensive, unreliable, very slow and limited. It costs approximately US\$ 99 plus shipping charges.

Lighthouse comes with a serial cable and AC adaptor. It has 4GB of internal storage with three USB 2.0 ports for connecting other USB devices, small micro-SD card slot to expand

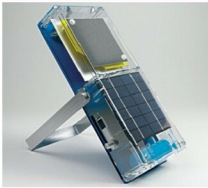


Fig. 2: Lantern is an anonymous portable library that constantly receives free data from Space (Image courtesy: www.indiegogo.com)



Fig. 3: Lighthouse receiver (Image courtesy: www.outernet.is)

storage, built-in Wi-Fi access point to add up to five devices and range up to 15.2m (50-feet). Ku/C-band satellite dish antenna, LNB and coaxial cable are also required but are not included in the kit.

Lantern. This is a small and portable device that can receive data from satellite networks directly without any satellite dish antenna. It has complete global reception capability. It is Wi-Fi-enabled and allows content to be accessed on other devices wirelessly. There is no limit to the amount of data that can be downloaded.

Lantern can be charged via a power adaptor and/or using solar power. Lantern receives data in small sizes of, say, 2MB per day (in future it could go up to 100MB to 200MB per day).

Lantern can also receive data using a dish antenna at high speed. Its battery works for 12 hours for data reception. It can even be used to charge devices such as smartphones.

Build-your-own receiver. Another way to access transmissions sent by the Outernet is to build a receiver that requires components like Raspberry Pi, satellite dish antenna, LNB and USB TV tuner card. After completion of the setup, the satellite dish receives transmissions sent by the Outernet, which are then sent to the USB TV tuner card and then to Raspberry Pi. Data received is stored on Raspberry Pi and can be accessed when a Wi-Fi dongle is connected to the Raspberry Pi and setup is done correctly. **EFY**

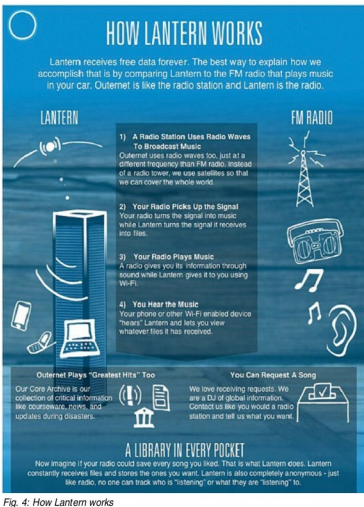


Fig. 4: How Lantern works